Harvard EEG Database (HEEDB v4.1) — data report and onboarding plan next to TUH

📄 **PDF version**

The Harvard Electroencephalography Database (HEEDB, Brain Data Science Platform, credentialed access) holds clinical EEG from Boston-area hospitals in BIDS format, with per-session **EEG report findings** and per-patient **health records** (ICD-10 diagnoses, ATC medications). Our brain-age and normal/abnormal models are trained on TUH (TUAB + TUEG cohorts). This notebook measures HEEDB with **the same analytics we ran on TUH**, so the two corpora can be compared column for column, and ends with the recommended data preparation and the plan to onboard HEEDB into the TUH pipeline (`docs/heedb_onboarding_plan.md`).

Everything is read **in place** through the BDSP S3 access point: the metadata CSVs, every EDF header (S3 byte-range reads, `dataset_maker/heedb_index.py`) and two-minute signal excerpts of a stratified sample (`labram/eval/heedb_analysis.py`). No HEEDB recording is copied for this report. **DUA:** HEEDB data and derivatives stay private (our copy is in the private bucket `s3://eeg-data-public/HEEDB/v4.1/`, never made public); this notebook shows only aggregates, no identifiers, and applies small-cell suppression: counts of 1–9 individuals are shown as `<10` and no statistic is computed on fewer than 10 (`heedb_analysis.suppress_small`).

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HEEDB: 341,084 index rows, 341,079 sessions, 72,124 adult routine sessions, 72,118 head
ers scanned | TUH: 71,348 recordings
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1. Sources and layout

HEEDB is organised by **site** (one hospital or partner each) and, within a site, by **service**: *Routine* (outpatient/inpatient EEG of 20–60 minutes, the study type of TUAB and most of TUEG), *LTM* (long-term ICU/inpatient monitoring, hours to days), *EMU* (epilepsy monitoring unit) and *OR* (intra-operative). Table 1.1 lists the sources the index joins; `dataset_maker/heedb_index.py::build_index` aligns them on `(site, patient, session)`.

Table 1.1. HEEDB sources joined into the index

One row per source table: what it holds and at which level (session = one EEG study, patient = all of a patient's records). TUH counterpart = the table that carries the same information in our TUH pipeline.

source	level	content	TUH counterpart
eeg-metadata/‹site›_eeg_metadata.csv	session	start, duration, service, age, sex, date of death	EDF header (age, sex, date) + folder
bids/‹site›/sub-*/ses-*/eeg/*_eeg.edf	record-ing	signal, 10-20 + auxiliary channels	TUEG / TUAB EDF
***_channels.tsv, *_eeg.json, *_annotations.csv**	record-ing	channel types, sfreq, events	— (EDF header only)
HEEDB_Metadata/‹site›_EEG_reports_findings.csv	session	normal/abnormal + ~45 findings	TUAB folder label; NEDC sheet `abnormal due to`, FIRDA…LPD
HEEDB_Metadata/HEEDB_ICD10_for_Neurology.csv	patient	ICD-10 codes in 18 neuro categories	NEDC sheet: 15 Y/N diagnoses
HEEDB_Metadata/HEEDB_Medication_ATC.csv	patient	ATC level-4 drug groups	NEDC sheet: medication list
HEEDB_Metadata/HEEDB_patients.csv	patient	sex, mean age, race, visit / ICD / drug counts	—

With d_i the duration of session i in seconds (metadata `DurationInSeconds`), the hours of a group G are

$$H(G) = \frac{1}{3600} \sum_{i \in G} d_i. \tag{1.1}$$

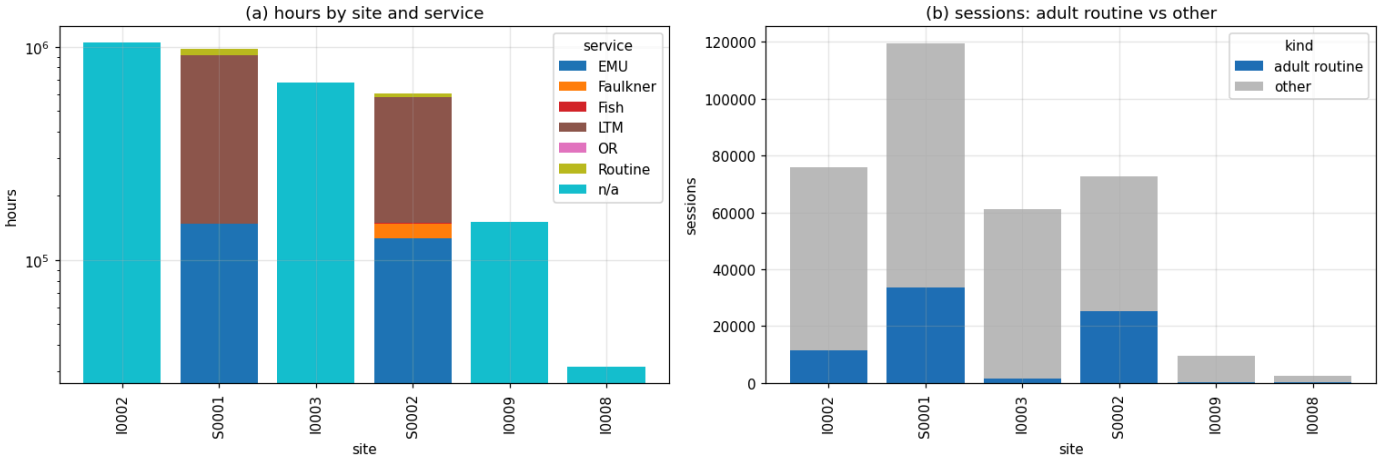
Table 1.2. Sessions, patients and hours per site and service

All sessions of the metadata tables. `n/a` = the site has no service label (I0002, I0003, I0008, I0009). Hours by Eq. (1.1); `median_min` = median session length in minutes.

site	service	sessions	patients	hours	median_min
I0002	**n/a**	75791	22113	1048582	809
I0003	**n/a**	61255	25045	681189	266
I0008	**n/a**	2501	530	31589	635
I0009	**n/a**	9485	2200	151716	1097
S0001	**EMU**	8943	1485	148076	1219
	LTM	61832	15005	766849	657
	OR	786	731	2925	204
	Routine	47887	35377	56099	67
S0002	**EMU**	7125	1275	126311	1371
	Faulkner	3760	3193	22436	36
	Fish	160	148	873	60
	LTM	32378	9185	429619	922
	Routine	29176	23044	26014	51

Figure 1.1. Where the HEEDB hours are

(a) Hours per site (Eq. (1.1)), split by service, log scale. (b) Sessions per site, split into adult routine (the onboarding set, age 18–89, Section 2) and everything else. All sessions.



2. Recording length

TUAB recordings are routine EEGs (median ≈ 20 min); TUEG adds a minority of long-term studies. HEEDB is dominated **in hours** by LTM/EMU monitoring (Figure 1.1(a)) but **in sessions** routine studies are the largest single group at S0001/S0002. A routine study is identified by the service label where the site has one, and otherwise by its length (`heedb_index.add_routine_flag`):

$$\text{routine}_i = \begin{cases} [\,\text{service}_i = \text{Routine}\,] \wedge [\,10 \le m_i \le 240\,] & \text{site with a service label} \\ [\,15 \le m_i \le 120\,] & \text{otherwise} \end{cases} \quad (2$$

where $m_i = d_i/60$ is the length in minutes and $[\cdot]$ is 1 when the condition holds. Quantiles Q_p in Table 2.1 are pandas' linear-interpolation sample quantiles of m_i.

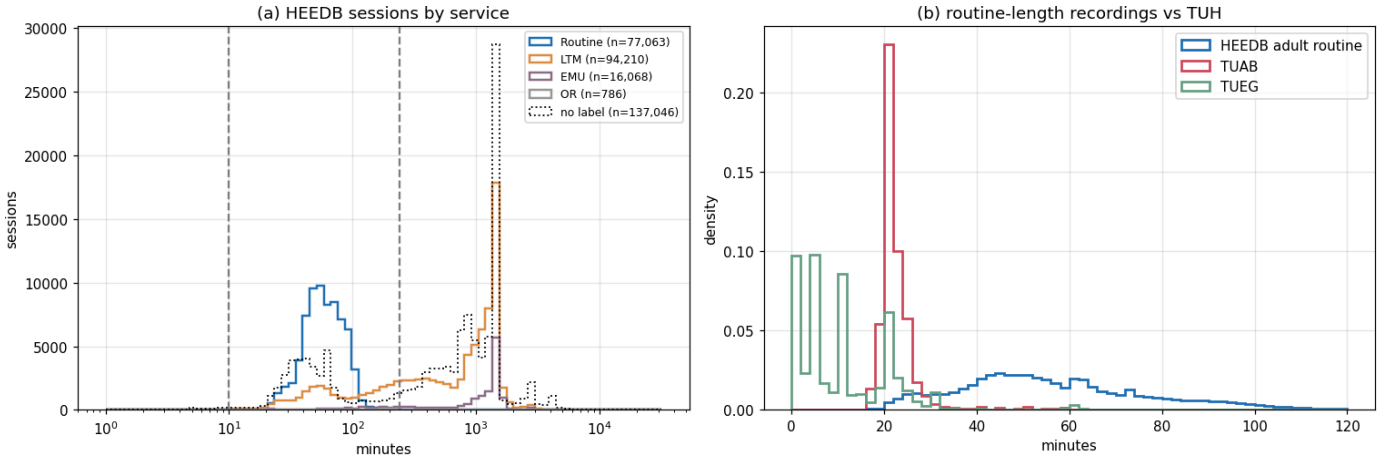
Table 2.1. Recording length (minutes) per group

Columns `p5 … p95` = Q_p of m_i; `n` = recordings with a known length. HEEDB rows are sessions (metadata duration); TUH rows are EDF files (header duration).

	p5	p25	p50	p75	p95	n
HEEDB Routine	28.6	45.0	57.4	76.3	102.7	72486.0
HEEDB LTM	35.6	200.6	778.1	1323.6	1497.3	91533.0
HEEDB EMU	53.6	535.8	1290.8	1442.9	1608.6	15543.0
HEEDB I0002/I0003 (no service label)	27.6	68.1	741.2	1432.5	1895.1	125727.0
HEEDB adult routine (Eq. 2.1)	26.2	42.1	53.5	69.2	97.3	72124.0
TUAB	19.2	20.1	21.1	23.4	28.4	2993.0
TUEG	0.1	4.8	10.0	20.1	56.5	68355.0

Figure 2.1. Distribution of recording length

(a) HEEDB sessions of the four main sites by service, minutes on a log axis; the dashed lines mark the 10–240 min routine window of Eq. (2.1). (b) Adult routine HEEDB sessions vs TUAB and TUEG recordings, 0–120 min, as density.



3. Demographics

Age is the session's `AgeAtVisit` (years, from the hospital record; I0008/I0009 from the date of birth). TUH's age comes from the EDF header, where TUH redacts ages ≥ 90 (`Age:999` , excluded, `labram/data/tuh_metadata.py`). HEEDB **does not redact**: it has ages above 89 and a few impossible values (negative, > 110), which the adult filter (18–89, the same range as `tuh_clinical.normative_mask`) removes. With a_i the age of recording i and $\mathrm{dec}(a) = 10\lfloor a/10 \rfloor$ capped at 90, the share of a group G in decade k is

$$s_k(G) = 100 \cdot \frac{|\{i \in G : \mathrm{dec}(a_i) = k\}|}{|\{i \in G : a_i \text{ known}\}|} \ \%. \tag{3.1}$$

Table 3.1. Overview: HEEDB next to TUH

Columns = groups (HEEDB all sessions of the four main sites, HEEDB adult routine, TUAB, TUEG, TUH adults). `hours` by Eq. (1.1), age in years (mean, sample SD), `female %` among recordings with a known sex, `with report label %` = share with a normal/abnormal call (HEEDB report, TUAB folder, NEDC sheet for TUEG), `abnormal %` among those (Eq. (4.1) without the decade restriction).

	HEEDB main sites	HEEDB adult routine	TUAB	TUEG	TUH adults 18-89
recordings	329093.0	72124.0	2993.0	68355.0	68587.0
patients	125642.0	58226.0	2329.0	13863.0	14069.0
hours	3308974.3	68654.4	1141.0	26028.3	25941.8
median minutes	340.6	53.5	21.1	10.0	10.0
age mean	44.0	54.0	49.1	51.1	52.0
age sd	27.9	19.8	17.4	18.1	17.0
female %	49.2	51.6	53.3	51.9	52.1
with report label %	52.1	50.7	100.0	84.4	84.4
abnormal % (of labelled)	72.0	60.3	49.2	56.1	55.7

Table 3.2. Age decade shares (%)

$s_k(G)$ of Eq. (3.1) per decade (rows) and group (columns). HEEDB all = all main-site sessions, including children and LTM.

	HEEDB all	HEEDB routine (all ages)	HEEDB adult routine	TUAB	TUEG
0–9	17.5	18.2	0.0	0.2	1.1
10–19	9.4	13.7	3.3	2.3	2.2
20–29	8.2	9.3	13.5	14.3	12.1
30–39	7.3	7.4	10.8	13.8	11.9
40–49	7.9	7.9	11.4	19.7	15.3
50–59	12.1	10.9	15.8	20.9	22.5
60–69	15.5	13.3	19.2	15.6	20.0
70–79	13.3	11.5	16.7	8.3	10.4
80–89	7.3	6.5	9.4	5.0	4.6
90+	1.5	1.3	0.0	0.0	0.0

Table 3.3. Age data quality

Per site: sessions with no age, a negative age, age over 89 (TUH would have redacted them), over 110 (implausible), and children (under 18). Counts.

	sessions	missing	negative	over 89	over 110	under 18
I0002	75791	0	<10	1959	<10	2275
I0003	61255	201	<10	<10	<10	57191
I0008	2501	0	0	46	0	11
I0009	9485	0	10	102	0	19
S0001	119448	44	0	1746	294	23409
S0002	72599	29	0	1147	96	96

Figure 3.1. Age distribution

(a) Age of routine sessions (all ages) per main site, 1-year bins, ages over 89 pooled in the last (90+) bin — I0003 is a paediatric hospital. (b) Adult routine HEEDB sessions vs TUAB and TUEG recordings, density; HEEDB ages > 89 are already removed by the adult filter.

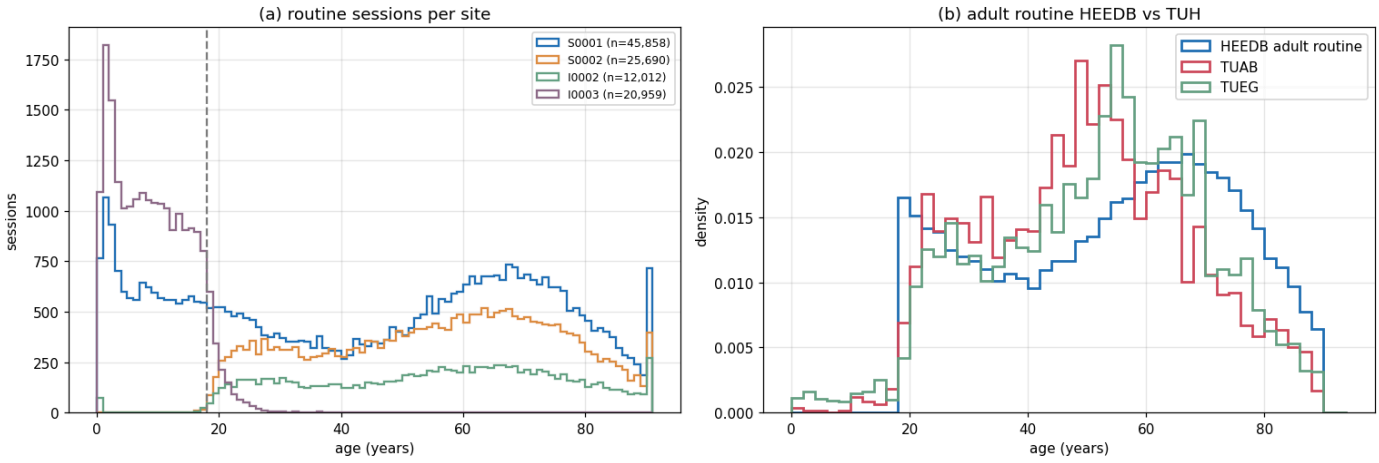
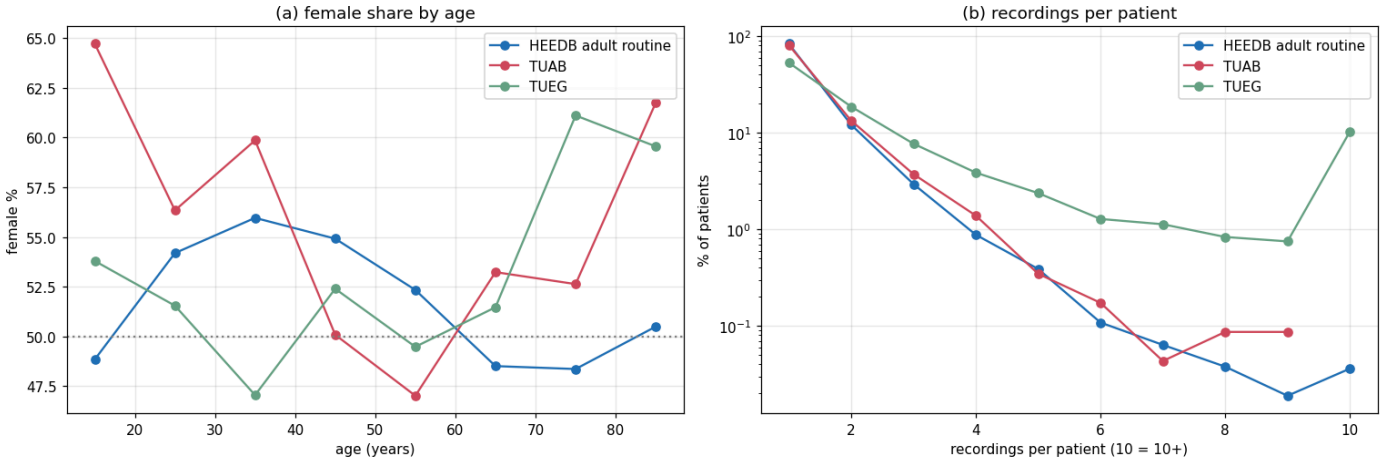


Figure 3.2. Sex balance and sessions per patient

(a) Share of female recordings per age decade (Eq. (3.1) groups), adult routine HEEDB vs TUAB and TUEG. (b) Sessions per patient in each corpus (HEEDB adult routine sessions; TUH recordings), share of patients, capped at 10.



4. EEG reports

Each HEEDB report was processed by BDSP into a normal/abnormal call and ~45 findings. A finding carries its **provenance**: `report` (stated in the clinical report), `annotation` (a Persyst/XLTEK machine or technician event in the recording) and `verified` (confirmed by an expert review). Only `report` / `verified` are comparable to TUH, whose labels come from reports (TUAB folders, the NEDC sheet's `abnormal due to`), so the index flags `find_<x>` use those two (`heedb_index.finding_flags`). The abnormal rate of a group G in age decade k is

$$r_k(G) = 100 \cdot \frac{|\{i \in G : \mathrm{dec}(a_i) = k,\ y_i = 1\}|}{|\{i \in G : \mathrm{dec}(a_i) = k,\ y_i \text{ known}\}|}\ \%, \tag{4.1}$$

with $y_i = 1$ for an abnormal report (TUAB: abnormal folder) and cells of fewer than 30 labelled recordings left empty. A finding's report prevalence is

$$p(x) = 100 \cdot \frac{|\{i : \mathrm{find}_x(i)\}|}{|\{i : y_i \text{ known}\}|}\ \%. \tag{4.2}$$

Table 4.1. Report coverage per site

Main-site sessions: `sessions` , sessions with a report (`with report`), of which normal / abnormal; the same for adult routine sessions. Counts and the abnormal share (%) of the reported adult routine sessions.

	sessions	with report	adult routine	adult routine with report	normal	abnormal	abnormal %
I0002	75791	9453	11634	677	312	365	53.9
I0003	61255	34453	1605	559	291	268	47.9
S0001	119448	88517	33628	25024	10685	14339	57.3
S0002	72599	38902	25257	10273	3199	7074	68.9
total	329093	171325	72124	36533	14487	22046	60.3

Figure 4.1. Abnormal rate by age

(a) r_k of Eq. (4.1) per age decade: HEEDB adult routine (report), HEEDB LTM (report), TUAB (folder label) and TUEG (NEDC sheet). (b) HEEDB adult routine per site.

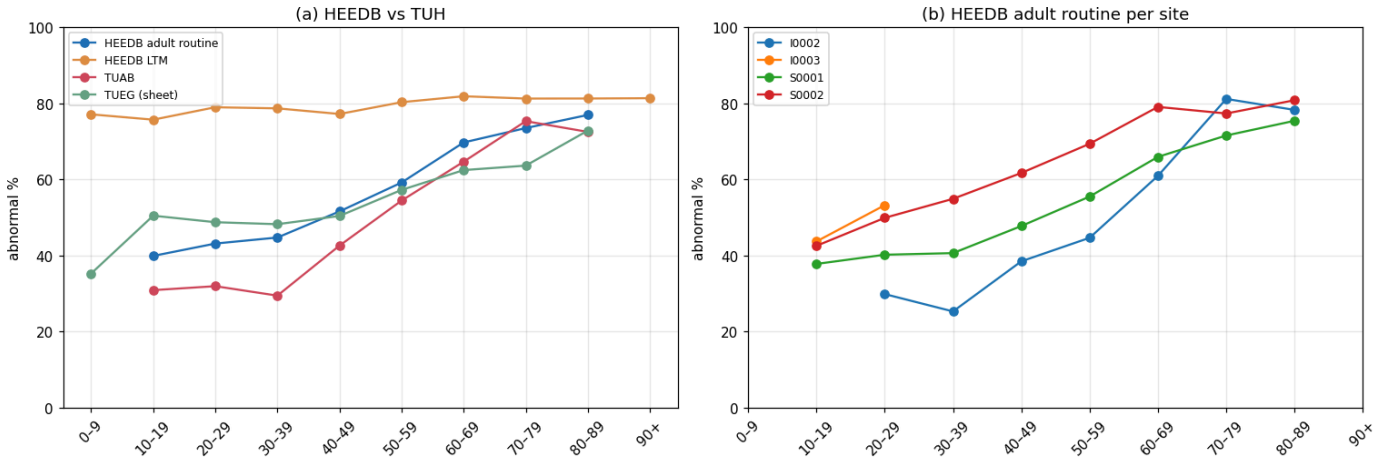


Table 4.2. Report findings in adult routine sessions

One row per finding, sorted by report mentions. `report`, `annotation`, `verified` = sessions whose provenance string contains that source (a session can count in several); `report %` = $p(x)$ of Eq. (4.2) over reported adult routine sessions; `mean age` (years) and `abnormal %` of the sessions with the finding (report/verified). `group` = candidate target group (`heedb_index.FINDING_GROUPS`).

finding	group	report	annotation	verified	report %	mean age	abnormal %
n1	sleep	30669	15865	0	83.9	53.3	55.5
pdr	other	25994	10578	1360	71.7	52.9	52.5
gen_slowing	generalized_slowing	24999	10672	2036	69.3	56.8	74.5
n2	sleep	22405	8820	0	61.3	52.1	54.8
awake	other	20885	19781	0	57.2	52.4	48.2
vertex_wave	sleep	13692	8641	753	37.6	48.9	47.1
spindles	sleep	13218	8532	716	36.3	48.8	47.0
spikes	epileptiform	10210	14098	0	27.9	53.4	80.2
k_complexes	sleep	8462	3688	0	23.2	48.7	43.5
seizure	seizure	5121	289	114	14.2	54.1	65.1
foc_slowing	focal_slowing	4696	240	921	14.4	57.6	89.6
diffuse_beta	other	2548	0	0	7.0	52.9	59.3
posts	normal_variants	2082	1937	530	6.3	46.2	34.3
grda	generalized_slowing	1874	1864	514	5.6	61.7	94.9
breach	normal_variants	1232	1261	0	3.4	55.5	92.9
gpd	epileptiform	875	506	310	2.8	66.3	95.8
bs	burst_suppression_low_voltage	735	1232	34	2.1	60.5	92.5
lpd	epileptiform	655	383	254	2.2	64.4	95.9
status	seizure	478	51	0	1.3	57.1	91.8
lrda	focal_slowing	454	426	165	1.5	59.9	95.7
ppr	epileptiform	294	12	0	0.8	41.4	50.7
low_voltage	burst_suppression_low_voltage	198	15	0	0.5	53.9	77.3
uninterpretable	other	159	0	0	0.4	57.5	74.8
jme	other	109	0	0	0.3	28.9	54.1
bets	normal_variants	93	182	57	0.4	52.6	40.0
jae	other	22	0	0	0.1	31.6	72.7
cjd	other	16	<10	0	0.0	64.7	62.5
bects	epileptiform	13	<10	0	0.0	27.9	23.1
bipd	epileptiform	<10	0	0	0.0	–	–
eses	epileptiform	<10	0	0	0.0	–	–
angelman	other	<10	0	0	0.0	–	–
jeavons	other	0	0	0	0.0	–	–
fold	other	0	<10	0	0.0	–	–
dravet	other	0	0	0	0.0	–	–
sunflower	other	0	0	0	0.0	–	–
wicket	normal_variants	0	393	144	0.4	56.5	40.3
wham	other	0	0	0	0.0	–	–

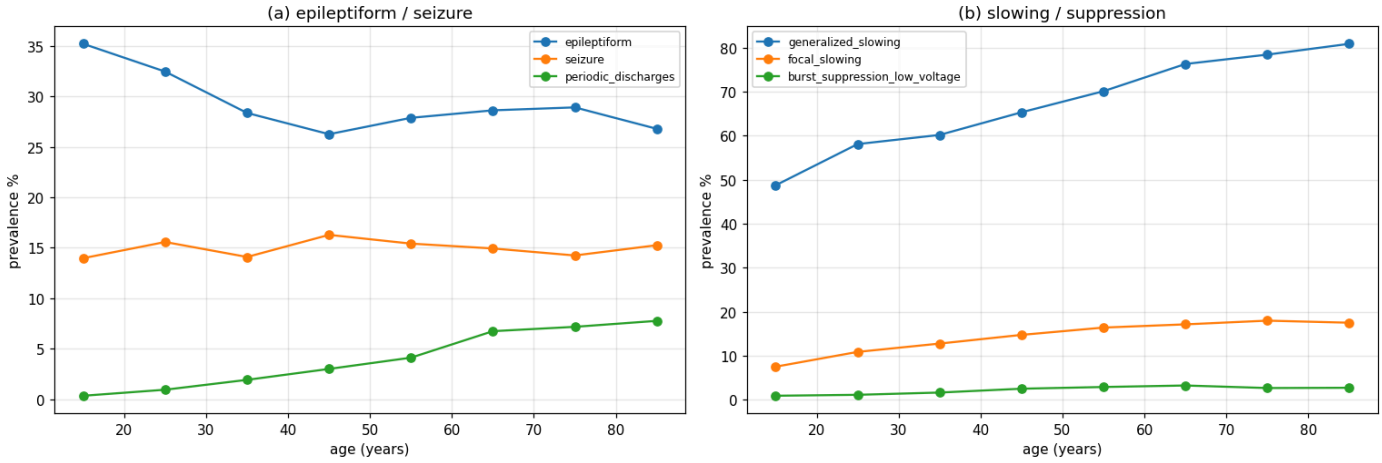
Table 4.3. Finding groups as candidate targets

Per group of `heedb_index.FINDING_GROUPS` over reported adult routine sessions: `n` positive sessions, prevalence (Eq. (4.2) on the group flag), `patients`, mean age and abnormal % of the positives; `normal-report positives` = positives whose report is normal (expected for sleep and normal variants, a warning sign for pathology groups).

	n	prevalence %	patients	mean age	abnormal %	normal-report positives
epileptiform	10534	28.8	8843	53.4	79.4	2167
seizure	5486	15.0	4900	54.2	66.4	1844
generalized_slowing	25558	70.0	21196	56.9	74.6	6483
focal_slowing	5606	15.3	4929	57.8	90.0	561
periodic_discharges	1684	4.6	1474	65.2	95.9	69
burst_suppression_low_voltage	892	2.4	819	59.5	89.0	98
normal_variants	3697	10.1	3462	49.8	53.7	1710
sleep	31869	87.2	27108	53.5	56.6	13816

Figure 4.2. Pathology finding groups by age

Prevalence (Eq. (4.2)) of the pathology groups per age decade among reported adult routine sessions. (a) Epileptiform, seizure, periodic discharges. (b) Generalized and focal slowing, burst suppression / low voltage.



5. Health records, aligned with the TUH clinical sheet

For TUEG, NEDC distributes one clinical row per session (15 Y/N diagnoses, a medication list, the abnormal finding, FIRDA/OIRDA/TIRDA/GPD/LPD), cleaned by `dataset_maker/tuh_clinical.py`. HEEDB gives **per-patient** ICD-10 codes and ATC drug groups instead. To compare the corpora — and to reuse `tuh_clinical.normative_mask` / `report` unchanged — the index rebuilds every sheet column from HEEDB sources (`heedb_index.tuh_aligned_flags`). Table 5.1 is the crosswalk. Two rules carry over from TUH: a patient **without** any ICD (or medication) record gets **unknown** (NA), not "healthy"; and `any_neuro_dx` / `sedated` are True if any member flag is True. For a flag c and the recordings K_c where it is known, Table 5.2 reports

$$\mathrm{prev}(c) = 100 \cdot \frac{|\{i \in K_c : c_i\}|}{|K_c|} \ \%, \tag{5.1}$$

$$\Delta a(c) = \bar{a}_{\{i \in K_c:\, c_i\}} - \bar{a}_{\{i \in K_c:\, \neg c_i\}} \quad (\text{years}), \tag{5.2}$$

$$\mathrm{abn}(c) = 100 \cdot \frac{|\{i \in K_c : c_i,\ y_i = 1\}|}{|\{i \in K_c : c_i,\ y_i \text{ known}\}|} \ \% \tag{5.3}$$

(`labram/eval/heedb_analysis.py::condition_table`; values with fewer than 20 positives are left empty).

Table 5.1. Crosswalk: TUH clinical columns rebuilt from HEEDB

One row per column of the cleaned NEDC sheet: kind, HEEDB source (ICD-10 / ATC prefixes, report finding) and whether the match is exact, approximate or missing.

TUH column	kind	HEEDB source	match
dx_electrographic_seizure	diagnosis	report finding `seizure` (report/verified)	exact
dx_clinical_seizure	diagnosis	ICD-10 R56, G40, G41	approx. (per patient, undated)
dx_stroke	diagnosis	ICD-10 I60, I61, I62, I63, I64, I69	approx. (per patient, undated)
dx_epilepsy	diagnosis	ICD-10 G40, G41	approx. (per patient, undated)
dx_parkinsons	diagnosis	ICD-10 G20, G21	approx. (per patient, undated)
dx_alzheimers	diagnosis	ICD-10 G30	approx. (per patient, undated)
dx_tbi	diagnosis	ICD-10 S06	approx. (per patient, undated)
dx_schizophrenia	diagnosis	ICD-10 F20, F25	approx. (per patient, undated)
dx_dementia	diagnosis	ICD-10 F01, F02, F03, G30, G31.0, G31.1, G31.83	approx. (per patient, undated)
dx_substance	diagnosis	ICD-10 F10, F11, F12, F13, F14, F15, F16, F17, F18, F19	approx. (per patient, undated)
dx_anxiety	diagnosis	ICD-10 F41	approx. (per patient, undated)
dx_depression	diagnosis	ICD-10 F32, F33	approx. (per patient, undated)
dx_bipolar	diagnosis	ICD-10 F31	approx. (per patient, undated)
dx_adhd	diagnosis	ICD-10 F90	approx. (per patient, undated)
dx_ms	diagnosis	ICD-10 G35	approx. (per patient, undated)
f_gpd	EEG finding	report finding gpd	exact
f_lpd	EEG finding	report finding lpd	exact
f_firda	EEG finding	report finding grda	approx. (no frontal/temporal split)
f_tirda	EEG finding	report finding lrda	approx. (no frontal/temporal split)
f_oirda	EEG finding	—	missing
med_anesthetic	medication	ATC N01A, N05CM	approx. (ever prescribed, not at EEG)
med_benzodiazepine	medication	ATC N05BA, N05CD, N03AE	approx. (ever prescribed, not at EEG)
med_barbiturate	medication	ATC N03AA, N05CA, N01AF	approx. (ever prescribed, not at EEG)
med_opioid	medication	ATC N02A, N01AH, N07BC	approx. (ever prescribed, not at EEG)
med_antiseizure	medication	ATC N03A	approx. (ever prescribed, not at EEG)
med_antipsychotic	medication	ATC N05A	approx. (ever prescribed, not at EEG)
abnormal_report	report	report `abnormal` (vs `normal`)	exact
age	demographics	AgeAtVisit (session)	exact
sex	demographics	SexDSC	exact

Table 5.2. Clinical flags: HEEDB adult routine vs TUEG

Per TUH clinical column (rows) and corpus (column groups): `known` recordings, `n` positives, prevalence (Eq. (5.1)), mean age, age vs rest (Eq. (5.2)) and abnormal % (Eq. (5.3)). HEEDB = adult routine sessions; TUEG = adult (18–89) recordings in the NEDC sheet.

	HEEDB adult routine						TUEG adults					
flag	known	n	prevalence %	mean age	age vs rest	abnor-mal %	known	n	prevalence %	mean age	age vs rest	abnormal %
dx_electrographic_seizure	36533	5204	14.2	54.1	-0.3	65.1	55368	2958	5.3	55.3	3.4	89.2
dx_clinical_seizure	30925	15502	50.1	49.9	-6.5	66.6	55368	6795	12.3	49.0	-3.6	57.9
dx_stroke	30925	5549	17.9	65.2	14.7	75.8	55368	2422	4.4	62.3	10.6	84.4
dx_epilepsy	30925	15429	49.9	49.9	-6.5	66.7	55368	4500	8.1	47.6	-4.9	56.7
dx_parkinsons	30925	937	3.0	71.8	19.3	72.5	55368	<10	0.0	–	–	–
dx_alzheimers	30925	920	3.0	74.8	22.4	74.0	55368	160	0.3	68.3	16.2	98.1
dx_tbi	30925	0	0.0	–	–	–	55368	653	1.2	50.3	-1.9	77.6
dx_dementia	30925	3378	10.9	72.0	21.2	74.8	55368	282	0.5	72.4	20.3	78.4
dx_ms	30925	583	1.9	53.2	0.0	58.6	55368	98	0.2	47.2	-4.9	44.9
dx_depression	30925	8771	28.4	55.4	3.1	61.1	55368	210	0.4	44.7	-7.4	36.2
dx_anxiety	30925	0	0.0	–	–	–	55368	181	0.3	40.5	-11.7	39.8
dx_bipolar	30925	0	0.0	–	–	–	55368	443	0.8	44.2	-8.0	54.4
dx_schizophrenia	30925	0	0.0	–	–	–	55368	189	0.3	52.9	0.8	32.3
dx_substance	30925	0	0.0	–	–	–	55368	1921	3.5	46.3	-6.0	73.7
dx_adhd	30925	1296	4.2	38.1	-15.7	46.2	55368	50	0.1	32.7	-19.5	52.0
f_gpd	36533	1012	2.8	66.3	12.2	95.8	55368	254	0.5	57.3	5.2	100.0
f_lpd	36533	800	2.2	64.4	10.3	95.9	55368	1612	2.9	67.3	15.6	100.0
f_firda	36533	2063	5.6	61.7	7.8	94.9	55368	632	1.1	51.8	-0.4	100.0
f_tirda	36533	534	1.5	59.9	5.6	95.7	55368	555	1.0	43.7	-8.5	100.0
f_oirda	0	0	0.0	–	–	–	55368	<10	0.0	–	–	–
med_anesthetic	50765	35497	69.9	55.9	8.4	62.0	55368	5425	9.8	58.4	7.0	55.4
med_benzodiazepine	50765	36290	71.5	54.2	3.0	61.7	55368	15621	28.2	52.6	0.6	60.0
med_barbiturate	50765	3266	6.4	54.5	1.2	65.2	55368	3180	5.7	51.9	-0.3	65.0
med_opioid	50765	33158	65.3	56.8	10.1	64.0	55368	7347	13.3	55.6	4.0	50.4
med_antiseizure	50765	33772	66.5	52.4	-2.7	65.6	55368	31507	56.9	50.0	-5.1	60.2
med_antipsychotic	50765	26477	52.2	56.9	7.4	65.8	55368	2456	4.4	49.2	-3.1	56.4
any_neuro_dx	33349	23287	69.8	54.2	2.8	65.8	55368	11150	20.1	51.7	-0.6	65.8
sedated	50765	43052	84.8	54.6	8.0	60.7	55368	21642	39.1	53.3	1.8	60.6

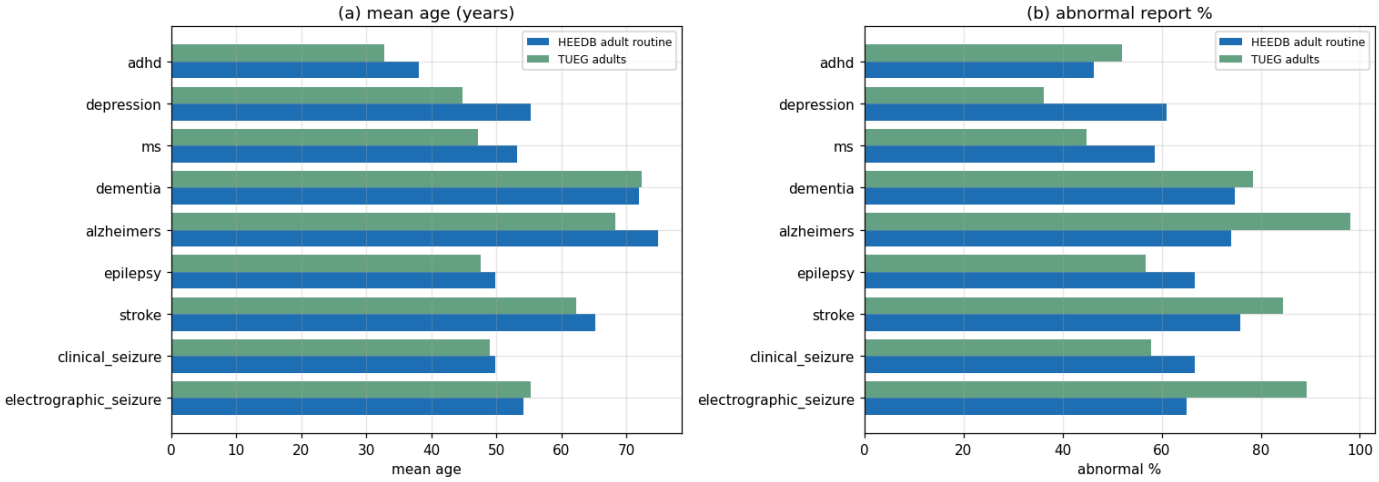
Table 5.3. HEEDB ICD-10 categories (patient level)

The 18 neurology categories of `HEEDB_ICD10_for_Neurology.csv` (HEEDB only; TUH has no ICD codes), over adult routine sessions of patients with at least one ICD record: positives, prevalence (Eq. (5.1)), mean age, age vs rest (Eq. (5.2)), abnormal % (Eq. (5.3)).

	known	n	prevalence %	mean age	age vs rest	abnormal %
miscellaneous	30925	26146	84.5	55.7	17.0	61.8
seizure_disorders	30925	15502	50.1	49.9	-6.5	66.6
headache_disorders	30925	12291	39.7	54.5	2.3	61.3
cerebral_degeneration	30925	10220	33.0	64.6	17.2	74.7
sleep_disorders	30925	9368	30.3	57.8	6.7	60.8
cerebral_lobe_dysfunction	30925	7403	23.9	62.4	12.2	68.2
cerebrovascular_diseases	30925	6986	22.6	65.1	15.5	71.4
movement_and_cerebellar_disorders	30925	6659	21.5	58.4	6.7	65.1
other_neurological_disorders	30925	4283	13.8	59.1	6.9	76.8
behavioral_cognitive_syndromes	30925	3632	11.7	48.7	-5.0	60.0
intracranial_and_spinal_tumors	30925	3026	9.8	57.6	5.0	78.7
peripheral_nervous_system_disorders	30925	1771	5.7	58.8	6.0	53.1
infections	30925	1506	4.9	57.0	4.1	77.6
developmental_delay	30925	1328	4.3	35.9	-18.0	62.7
muscular_dystrophies_and_other_myopathies	30925	368	1.2	53.4	0.2	67.6
motor_neuron_diseases	30925	149	0.5	62.4	9.3	64.6
cranial_nerve_disorders	30925	125	0.4	56.9	3.8	66.7
genetic_disorders	30925	70	0.2	44.9	-8.2	63.9

Figure 5.1. Age and abnormal rate per condition, HEEDB vs TUEG

(a) Mean age of recordings with each diagnosis (Table 5.2), HEEDB adult routine vs TUEG adults. (b) Abnormal rate (Eq. (5.3)) for the same diagnoses. Only flags with ≥ 20 positives in both.



6. Cohorts for brain age

Our TUH brain-age cohorts start from `tuh_clinical.normative_mask`: in the sheet, report-normal, no neurological diagnosis, no sedating drug, age 18–89. Applied unchanged to HEEDB (the flags of Table 5.1), the **medication** term behaves differently: HEEDB medications are every drug group ever prescribed to the patient, not the drugs at the time of the EEG, so most hospital patients are "sedated" (Table 5.2). And `any_neuro_dx` is only known for patients with an ICD record, so requiring it to be False drops every patient without a health record. Table 6.1 therefore counts four nested rules; "no *known* neuro dx" (report-normal, no neurological ICD code or seizure finding, patients without a record allowed) is the HEEDB analogue of the TUH healthy cohort.

Table 6.1. Normative candidates per age decade

Adult routine sessions per decade: report-normal; + no known neurological diagnosis (`any_neuro_dx` not True); + no neurological diagnosis with an ICD record (`any_neuro_dx` False);

- not sedated (the full TUH rule). TUEG column = the full TUH rule on TUEG adults, for scale. Counts.

	report-normal, 18–89	+ no known neuro dx	+ no neuro dx (ICD record)	+ not sedated (TUH rule)	TUEG (TUH rule)
10–19	719	432	256	69	199
20–29	2806	1465	888	162	2170
30–39	2112	1165	702	89	2006
40–49	1845	998	574	51	2492
50–59	2316	1240	742	56	2914
60–69	2200	1133	659	36	2004
70–79	1695	835	510	33	1029
80–89	794	364	183	15	342
total	14487	7632	4514	511	13156

7. The EEG signal: headers

The header of every adult routine EDF was read by an S3 range request (`heedb_index.fetch_header` / `header_row`; 256 bytes + 256 per signal) — the same fields the TUH scan records (`tuh_metadata.recording_row`) plus units, prefilter and amplitude resolution. The digital resolution of a channel with physical range $[P_-, P_+]$ (µV) stored in the digital range $[D_-, D_+]$ is

$$\mathrm{LSB} = \frac{P_+ - P_-}{D_+ - D_-} \quad \mu\mathrm{V\ per\ digital\ unit.} \tag{7.1}$$

HEEDB channel labels are bare BIDS names (`Fp1`, `T7`); `tuh_metadata.bids_canonical_channel` maps them onto the 23 TUH channels (the 10-10 names T7/T8/P7/P8 become T3/T4/T5/T6).

Table 7.1. Header facts of adult routine EDFs vs TUH

Share (%) of recordings per value: native sampling rate (Hz), EDF type, physical unit, number of the 23 TUH channels present, and the median LSB (Eq. (7.1)). HEEDB = adult routine EDFs with a scanned header; TUH = all TUAB + TUEG recordings (`tuh_index`; units and LSB not in that index).

	HEEDB adult routine	TUH (TUAB + TUEG)
recordings	72,118	71,348
sampling rate (Hz)	200: 68.3%; 256: 20.8%; 512: 10.8%; 1024: 0.1%; 4096: 0.0%; 2048: 0.0%	256: 51.3%; 250: 37.4%; 1000: 4.4%; 400: 4.4%; 512: 2.4%
EDF type	EDF+C: 100.0%	EDF
unit	uV: 100.0%	uV (TUH QC sample)
TUH channels present (of 23)	19: 70.0%; 21: 29.9%; 2: 0.1%; 13: 0.0%	23: 78.1%; 21: 17.4%; 19: 4.4%; 17: 0.1%; 22: 0.0%; 0: 0.0%
no 10-20 channel found	0 recordings	–
signals per file	median 31 (p5 23, p95 51)	median 33 (p5 28, p95 41)
median LSB (µV, Eq. 7.1)	0.266	–
prefilter field	(empty): 66.0%; None: 34.0%	HP/LP/N fields, mostly placeholders
header vs metadata length	median |Δ| 1 s	–

Figure 7.1. Channel coverage

(a) Share (%) of recordings in which each of the 23 TUH channels is present (bare BIDS label mapped by `bids_canonical_channel`), HEEDB adult routine vs TUH. (b) Auxiliary channel types in HEEDB adult routine EDFs (`heedb_index.AUX_PATTERNS`, % of recordings).

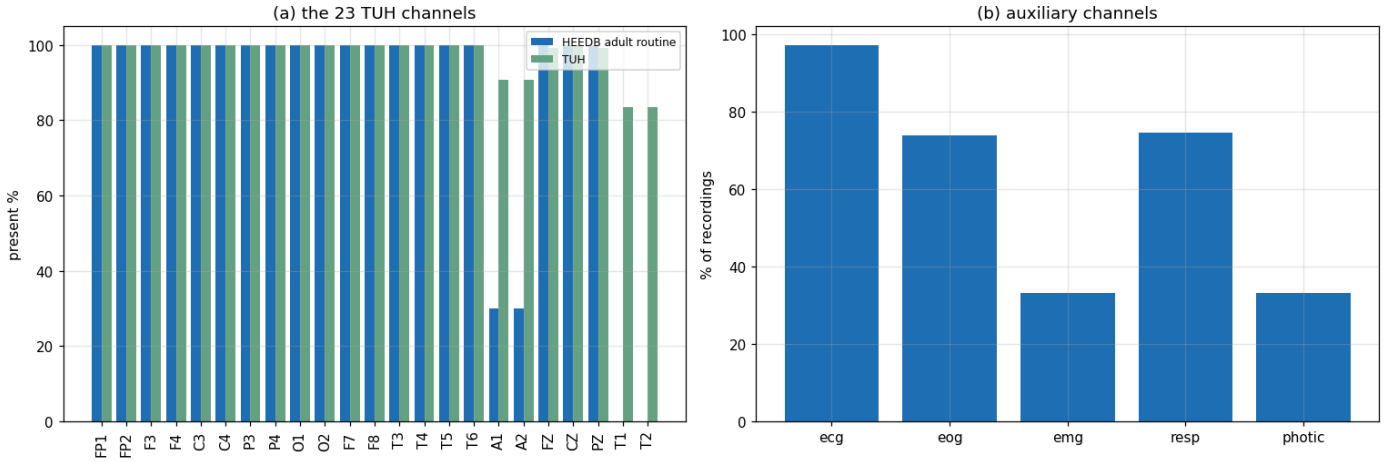


Table 7.2. Sampling rate per site

Adult routine EDFs with a scanned header: share (%) of each native sampling rate (Hz) per site, and the number of headers. Cells with fewer than 10 recordings show `<10 rec.` (small-cell suppression).

sfreq site	200	256	500	512	1024	2048	4096	16384	headers
I0002	11.1	80.2	0.0	7.8	0.5	0.1	0.2	<10 rec.	11634
I0003	7.2	66.4	0.0	26.2	<10 rec.	<10 rec.	0.0	0.0	1605
S0001	73.9	7.0	<10 rec.	19.1	0.0	<10 rec.	0.0	0.0	33621
S0002	91.1	8.9	0.0	0.0	<10 rec.	0.0	0.0	0.0	25258

8. The EEG signal: content and line noise

The header says nothing about the signal itself. A **stratified sample** of adult routine EDFs — up to 30 per site × report label (normal, abnormal, unlabelled), one per patient (`heedb_analysis.stratified_sample`) — was measured exactly like the TUH QC sample of `notebooks/tuh_preprocessing_analysis.ipynb` (`preprocessing_qc.analyze_recording`, 187 TUAB/TUEG recordings). One difference: many HEEDB files contain long **flat** stretches (amplifier on, no signal) and non-physiological stretches (electrodes not yet attached), so each recording is first probed at 12 evenly spaced 10 s excerpts (`heedb_analysis.signal_profile`). With $\sigma_{c,j}$ the standard deviation (µV) of channel c in probe j and C the TUH channels present,

$$\begin{aligned} \mathrm{flat}_j &= \left[\frac{1}{|C|}\sum_{c\in C}[\sigma_{c,j} < 0.5\,\mu\mathrm{V}] \geq 0.5\right], \\ \text{flat time} &= \tfrac{1}{12}\textstyle\sum_{j=1}^{12}\mathrm{flat}_j, \end{aligned} \tag{8.1}$$

$$\begin{aligned} \mathrm{noisy}_j &= \neg\mathrm{flat}_j \wedge \left[\,\mathrm{median}_c\,\sigma_{c,j} \notin [2, 150]\,\mu\mathrm{V}\right], \\ \text{noisy time} &= \tfrac{1}{12}\textstyle\sum_{j=1}^{12}\mathrm{noisy}_j. \end{aligned} \tag{8.2}$$

The spectral measures use 120 s of non-flat data records starting at the middle physiological probe. The line-noise prominence of a frequency f_0 in the channel-median Welch PSD P (4 s

Hann, 50 % overlap) is (`preprocessing_qc.line_prominence_db`)

$$\mathrm{LNP}(f_0) = 10\log_{10}\frac{\operatorname{mean} P\big([f_0-0.5,\ f_0+0.5]\big)}{\operatorname{median} P\big([f_0-5,\ f_0-2]\cup[f_0+2,\ f_0+5]\big)}\ \ \mathrm{dB}, \qquad (8.3)$$

~0 dB = no line. 76 Hz is where the 180 Hz harmonic folds to when sampled at 256 Hz ($256-180=76$). Amplitude measures (`preprocessing_qc.amplitude_stats`): median and 99th percentile of $|x|$ (µV), `clip_frac` = share of samples at ≥ 99.5 % of the physical maximum, `flat_channel_frac` = share of channels flat in some 10 s window, `dc_offset_uv` = median over channels of $|\overline{x}_c|$ (µV).

`HEEDB sample: 360 EDFs, 358 analysed, 2 without a physiological probe | TUH QC sample: 187`

Table 8.1. Signal integrity, HEEDB sample vs TUH QC sample

Median (and share of recordings where relevant) of: flat time (Eq. (8.1)), noisy time (Eq. (8.2)), recordings flat ≥ 25 % of the time, time of the first non-flat probe (s), median and 99th-percentile $|x|$ (µV), clip fraction, flat-channel fraction and DC offset (µV) of the analysed 120 s. TUH has no flat/noisy probes (its QC used 60–180 s directly).

	HEEDB sample (all)	HEEDB I0002	HEEDB I0003	HEEDB S0001	HEEDB S0002	TUAB QC	TUEG QC
recordings	360.0	90.0	90.0	90.0	90.0	80.0	107.0
median flat time %	16.7	0.0	0.0	33.3	41.7	–	–
mean flat time %	21.1	2.5	10.6	27.5	43.6	–	–
recordings flat ≥ 25 % (%)	46.7	2.2	21.1	70.0	93.3	–	–
mean noisy time %	10.1	7.3	5.8	15.2	11.9	–	–
median first signal (s)	0.0	0.0	0.0	0.0	0.0	–	–
median |x| (µV)	22.2	20.6	23.1	23.2	23.0	8.7	15.3
p99 |x| (µV)	114.7	101.2	108.2	138.7	129.5	80.0	124.5
clip frac (%)	0.7	0.0	0.0	0.2	2.8	0.0	0.1
flat channel frac (%)	0.0	0.0	0.0	0.0	0.1	0.0	0.0
DC offset (µV)	15.4	15.9	17.4	14.1	15.1	0.4	7.4

Figure 8.1. Flat signal inside HEEDB recordings

(a) Flat time per recording (Eq. (8.1)), density over the sample, per site. (b) State of each of the 12 probes by its position in the recording (probe 1 = start, 12 = last 10 s): share (%) of recordings in which the probe is flat (Eq. (8.1)), noisy (Eq. (8.2)) or physiological signal.

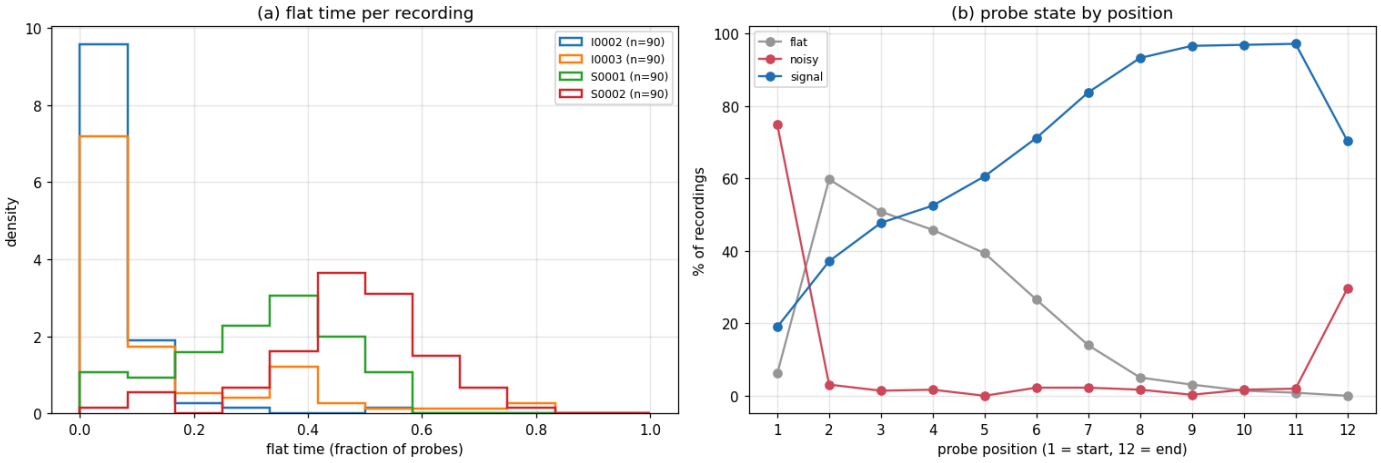


Table 8.2. Mains and harmonic lines in the raw signal

Per corpus: median LNP (Eq. (8.3), dB) of the raw channel-median PSD at 50, 60, 76 (180 Hz alias at 256 Hz) and 120 Hz, and the share (%) of recordings with LNP > 3 dB (a visible line).

	LNP 50 Hz (dB)	50 Hz line > 3 dB (%)	LNP 60 Hz (dB)	60 Hz line > 3 dB (%)	LNP 76 Hz (dB)	76 Hz line > 3 dB (%)	LNP 120 Hz (dB)	120 Hz line > 3 dB (%)
HEEDB sample	-0.0	0.0	11.5	83.2	0.3	12.0	0.2	5.9
TUAB QC	0.1	0.0	1.9	37.5	0.1	0.0	-0.3	6.2
TUEG QC	-0.0	0.0	4.7	60.7	0.0	0.9	0.7	27.1

Figure 8.2. Grand-average spectrum, raw

Median over recordings of the channel-median PSD (μV²/Hz, log scale) of the raw 120 s excerpts, interpolated onto a common 0.25 Hz grid. (a) 0.5–100 Hz, HEEDB sample vs TUAB and TUEG QC samples. (b) 40–100 Hz zoom with 50/60/76 Hz marked.

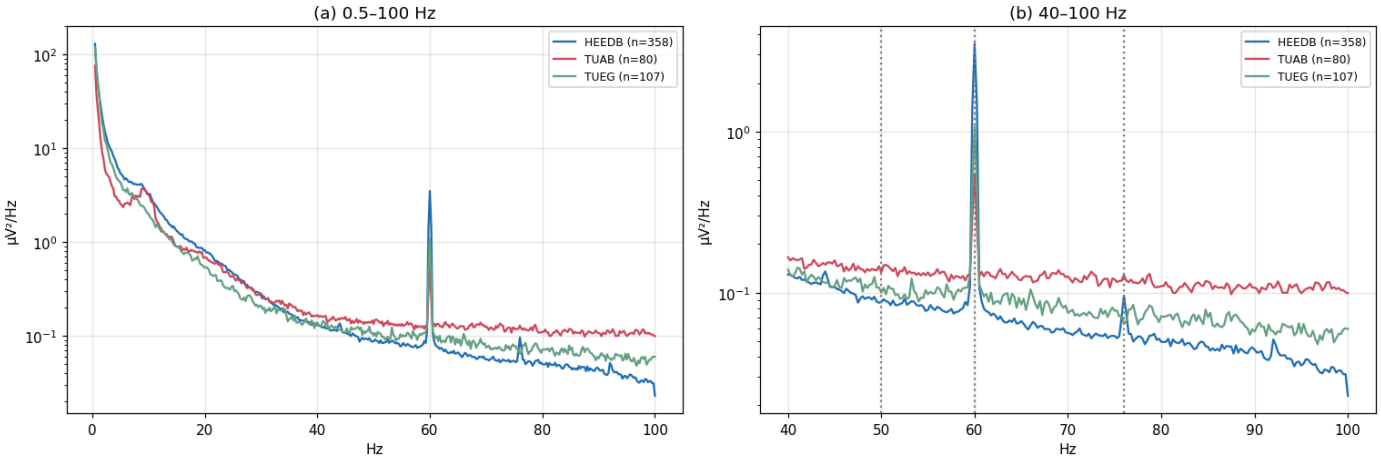


Table 8.3. Line left at the model input, per preprocessing protocol

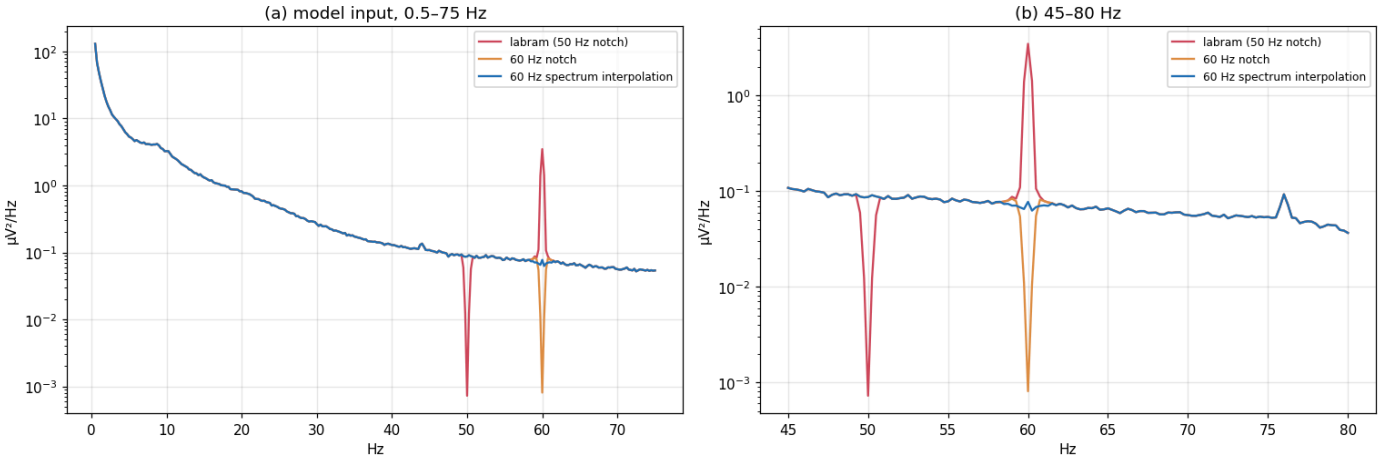
For each protocol of `preprocessing_qc` (band-pass 0.1–75 Hz, the named line removal, resample to 200 Hz): median LNP (Eq. (8.3), dB) after processing at 60 Hz and at 76 Hz, and the share (%) of 55–65 Hz power in the 0.5–75 Hz band (`band_fraction`), HEEDB sample vs

TUEG QC sample. LaBraM = the original 50 Hz notch; interpolation = `labram/data/line_removal.py::spectrum_interpolate`.

		LNP 60 Hz (dB)	LNP 76 Hz (dB)	55–65 Hz power share (%)
labram (50 Hz notch)	**HEEDB**	11.54	0.58	1.95
	TUEG	4.65	0.35	1.29
band-pass only (no line removal)	**HEEDB**	11.54	0.58	1.95
	TUEG	4.65	0.35	1.28
60 Hz notch	**HEEDB**	-5.27	0.58	0.37
	TUEG	-5.31	0.35	0.68
60 Hz spectrum interpolation	**HEEDB**	-0.37	0.58	0.40
	TUEG	-0.35	0.35	0.72
50 + 60 Hz spectrum interpolation	**HEEDB**	-0.37	0.58	0.40
	TUEG	-0.35	0.35	0.72
60 Hz notch, 45 Hz low-pass	**HEEDB**	-5.35	-4.76	0.00
	TUEG	-5.83	-5.78	0.00

Figure 8.3. Spectrum at the model input, HEEDB

Median channel-median PSD of the HEEDB sample after each protocol (output at 200 Hz, μV²/Hz, log). (a) 0.5–75 Hz. (b) 45–80 Hz zoom: the LaBraM recipe (50 Hz notch) leaves the 60 Hz line, a notch cuts a hole, interpolation removes the line without one.



9. Recommended data preparation and filtering

The TUH recipe (`dataset_maker/make_TUH_npy.py`, analysed in `notebooks/tuh_preprocessing_analysis.ipynb`) carries over to HEEDB with the changes Sections 7–8 call for. The output contract stays **identical** — per-recording `recordings/<stem>.npy` `[T, 23]` at 200 Hz in μV, whole 10 s windows, `manifest.json` with a per-recording channel mask — so `TUABAgeNpyLoader`, window QC, cohort building and training read HEEDB exactly like TUH.

Table 9.1. TUH recipe vs recommended HEEDB recipe

One row per step: what the TUH pipeline does, what HEEDB needs, and the evidence (table/figure of this notebook).

step	TUH (current)	HEEDB (recommended)	evidence
selection	TUAB/TUEG, age 1–89, ≥ 60 s	routine service (Eq. 2.1), age 18–89, 10–240 min; drop files with < 19 of the 23 channels (43 have 2)	Tables 2.1, 3.3, 7.1
channels	`EEG X-REF/LE` → 23 TUH channels, missing zero-filled + mask	bare BIDS labels via `bids_canonical_channel` (T7/T8/P7/P8 → T3/T4/T5/T6) → same 23 + mask; A1/A2 absent in 70 % and T1/T2 in all files → train with missing-channel handling (`channel_drop`, as the TUH `anymiss` cohorts)	Table 7.1, Figure 7.1(a)
reference	referential, CAR at train time (`labram_plus`)	referential; same CAR at train time (also removes the larger DC offset)	Tables 7.1, 8.1
band-pass	0.1–75 Hz zero-phase FIR (MNE)	same	Figure 8.2
line removal	spectrum interpolation at 50 + 60 Hz (`interp50-60`)	spectrum interpolation at 60 Hz (the TUEG recipe works unchanged; 50 Hz is a no-op); the 76 Hz alias of 180 Hz lies above the 75 Hz band edge	Tables 8.2–8.3, Figure 8.3
resampling	native 250–1000 Hz → 200 Hz (FFT)	most files are already 200 Hz (no resampling); 256/512 Hz → 200 Hz	Table 7.2
units / dtype	µV, float16 npy	µV (all headers in uV), float16 npy	Table 7.1
flat / dead signal	window QC `min_std` 0.5 µV (opt-in)	required: detect flat records at conversion time and drop flat 10 s windows (S0001/S0002 files are flat for long stretches in their first half); record per-window stats in `window_stats.npz`	Table 8.1, Figure 8.1
artifact windows	window QC `max_abs` 800 µV (opt-in)	on by default; also drop the first/last probe-length of non-physiological signal	Table 8.1, Figure 8.1(b)
amplitude scale	µV as recorded	µV as recorded; check per-site scale after filtering (raw median |x| 22 µV vs 9–15 µV in TUH, partly DC); per-patch z-scoring (`labram_plus`) neutralises it	Table 8.1
labels	age from EDF header; abnormal from TUAB folder / NEDC sheet	age from metadata (`AgeAtVisit`); abnormal from the report; findings only after validation (negations are counted); ICD/ATC flags per patient (Table 5.1)	Sections 4–5

10. Plan: onboarding HEEDB like TUH

Each TUH pipeline stage has a HEEDB counterpart (Table 10.1); most reuse the TUH code with a pluggable corpus adapter. The data move is a **single pass over all adult routine sessions** (Table 10.2), copied server-side from the BDSP access point into a private, KMS-encrypted bucket in our account (same region, no egress), then converted to npy on AWS Batch Spot. Full plan: `docs/heedb_onboarding_plan.md`.

Table 10.1. TUH pipeline stage → HEEDB counterpart

One row per stage: the TUH module, the HEEDB module and its state (done = in this change).

stage	TUH	HEEDB	state
download	download_tuh.py (rsync → s3 → verify)	download_heedb.py (access point → private S3, s5cmd)	next
header scan	tuh_metadata.scan_recordings	heedb_index.fetch_header / header_row (range reads)	done
index	tuh_index.build_index → tuh_index.parquet	heedb_index.build_index → heedb_index.parquet	done
clinical join	tuh_clinical.load_sheet / join_index	heedb_index.tuh_aligned_flags (Table 5.1)	done
data report	tuh_index.summarize, tuh_preprocessing_analysis.ipynb	this notebook	done
EDF → npy	make_TUH_npy.py convert/merge	make_TUH_npy.py + BIDS channel map, flat-window drop	next
batch conversion	scripts/submit_batch_npy.py	same, --corpus heedb, private bucket	next
window QC	labram.data.window_qc	same	reuse
patient split	tueg_patient_split.hash_split	same, namespace `heedb`, + held-out HEEDB test	next
cohorts	make_cohort_dataset.py	+ HEEDB selector, labels.parquet sidecar	next
training	TUAB_AGE npy loader, MultiTaskHead	+ source weights, masked multi-label heads	next

Table 10.2. The download set: adult routine sessions per site

Per main site: adult routine sessions (Eq. (2.1), age 18–89), patients, hours (Eq. (1.1)), sessions with a report (normal / abnormal), sessions whose EDF was found in the BIDS listing, and raw size (TB) from the listing. One pass, no waves.

site	sessions	patients	hours	normal	abnormal	EDFs found	raw TB
I0002	11634.0	9971.0	9353.8	312.0	365.0	11634.0	1.2
I0003	1605.0	1280.0	1230.7	291.0	268.0	1605.0	0.2
S0001	33628.0	26856.0	35702.2	10685.0	14339.0	33621.0	3.2
S0002	25257.0	20119.0	22367.7	3199.0	7074.0	25258.0	0.9
total	72124.0	58226.0	68654.4	14487.0	22046.0	72118.0	5.5

Table 10.3. Planned cohorts and training targets

One row per cohort to build after conversion: definition, size (adult routine sessions; from Tables 4.1 and 6.1), and its use.

cohort	definition	sessions	use
heedb_routine_v1	all adult routine sessions	72,124	brain age (all), abnormal + finding heads
heedb_normal_v1	report-normal, no known neuro dx (Table 6.1)	7,632	normative brain age
heedb_labelled_v1	adult routine with a report	36,533	normal/abnormal, findings
heedb_icd_v1	adult routine of patients with an ICD record	30,925	ICD-group heads (masked, down-weighted)
tuh_heedb_v1	TUH all_both_v3 + heedb_routine_v1, one private manifest	72,124 + TUH	joint training

Order of work. (1) Private bucket + one-pass copy of the Table 10.2 set (≈ 72k sessions), resumable, verified by size/ETag. (2) Generalise `make_TUH_npy.py` to the BIDS channel map and the flat-window rule of Table 9.1; convert a 2k pilot, check it against Section 8, then the full set on Batch Spot. (3) Patient-level split (patients that appear at two sites stay together) with a fixed held-out HEEDB test set; build the cohorts of Table 10.3. (4) Train: TUH-only baseline evaluated on HEEDB (transfer), HEEDB-only, TUH + HEEDB, then the abnormal / finding / ICD heads; every run scored on the TUAB eval set, the TUH cohort test set and the HEEDB test set per site.

11. Findings

Size and scope. HEEDB v4.1 has 341k recording sessions (≈ 3.3 M hours, ≈ 366 TB of EDF) from 128k patients, but most of the volume is long-term monitoring (Table 1.2, Figure 1.1). The part that matches our TUH task — **adult routine EEG, age 18–89** — is 72,124 sessions from 58,226 patients, ≈ 68.7k hours (Table 3.1), i.e. **24× TUAB and about as many recordings as all TUAB + TUEG**, and 2.5× the TUH hours since a HEEDB routine study runs a median 53.5 min vs 21 min (TUAB) and 10 min (TUEG) (Table 2.1). The download set (EDF + sidecars) is 376,751 objects, **5.48 TB**, in one pass (Table 10.2).

Demographics. Adult routine HEEDB is older than TUH (mean 54.0 vs 49.1 TUAB / 51.1 TUEG years, SD 19.8) with a much heavier 60–89 tail — 45.3 % of sessions vs 28.9 % TUAB — which is where our brain-age errors are largest (Table 3.2, Figure 3.1(b)); sex is balanced (51.6 % female). Ages are not redacted at 90: 5k sessions are over 89 and ≈ 400 over 110 (Table 3.3), so the 18–89 filter is required, and it keeps the label range identical to TUH. I0003 is a paediatric hospital (93 % of its sessions are children), so it adds little to the adult set.

Reports. Half of the adult routine sessions (36,533) have a report label, 60.3 % abnormal vs 49.2 % in TUAB and 56.1 % in TUEG (Tables 3.1, 4.1); the abnormal rate rises with age as in

TUH (Figure 4.1). That is **12× the labelled normal/abnormal recordings of TUAB**. The ~45 findings are an NLP extraction with provenance; mentions are counted even when negated — generalized slowing is flagged in 44.8 % and epileptiform activity in 15.0 % of *normal* reports (Tables 4.2, 4.3). The rare, specific findings (GPD, LPD, GRDA, LRDA, burst suppression, focal slowing) are ≈ 90 % abnormal or more and usable; the common ones need validation before they become training targets.

Health records. Every TUH clinical column except TBI, anxiety, bipolar, schizophrenia, substance use and OIRDA has a HEEDB counterpart (Table 5.1) — the ICD file is a neurology extract, so psychiatric codes beyond depression and ADHD are absent. The diagnoses behave like TUH's: stroke, dementia and Alzheimer patients are 15–22 years older than the rest and mostly abnormal, epilepsy patients younger (Table 5.2, Figure 5.1). Prevalences are far higher than in TUEG (epilepsy 49.9 % vs 8.1 %, stroke 17.9 % vs 4.4 %) because ICD codes are lifetime, per patient, and only exist for the 30,925 adult routine sessions of patients with a record. Medications are lifetime too (84.8 % "sedated"), so the TUH sedation rule cannot be applied as is (Table 6.1). The 18 HEEDB ICD categories (Table 5.3) give large weak-label groups for multi-task heads: seizure disorders 15.5k, cerebral degeneration 10.2k, cerebrovascular 7.0k sessions.

Normative cohort. Report-normal adult routine sessions number 14,487; excluding known neurological diagnoses leaves 7,632 (Table 6.1) — comparable to the 13,156 TUEG sessions that pass the TUH rule, and with more 60–89-year-olds.

Signal. All 72,118 adult routine headers are EDF+C in µV; 68.3 % are already sampled at 200 Hz (S0002 91 %), the rest 256/512 Hz (Tables 7.1–7.2). 70 % lack A1/A2 and all lack T1/T2, so 19 of the 23 TUH channels are the common set (Figure 7.1(a)). On the 358 analysed sample recordings the 60 Hz mains line is stronger and more frequent than in TUH (median LNP 11.5 dB, visible in 83 % vs 61 % TUEG / 38 % TUAB; Table 8.2); the TUEG recipe (60 Hz spectrum interpolation) removes it without a hole, while the original LaBraM 50 Hz notch leaves it untouched (Table 8.3, Figure 8.3). The new problem is **flat signal**: S0001/S0002 routine files are flat for long stretches in their first half (93 % of S0002 and 70 % of S0001 sample files are flat ≥ 25 % of probes; Table 8.1, Figure 8.1), so flat-window removal must be part of the conversion, not an option.

Recommendation. Onboard the 72k adult routine sessions with the TUH npy contract and the recipe of Table 9.1 — same channels/band-pass/line removal/200 Hz as TUEG, plus mandatory flat- and artifact-window removal and missing-ear-channel handling — then build the cohorts of Table 10.3. Split by patient across sites (1,057 patients appear at two sites).